

Guest Editorial: Introduction to the Special Section on Large Models for Future Vehicles and Transportation

COMBINING Large Models (LMs) with Artificial Intelligence (AI) technologies, edge computing and cloud computing, can create a powerful synergy that addresses the challenges and requirements of future vehicles and transportation systems. In light of this potential, this special section provides a venue to cover comprehensively algorithms, frameworks, technologies, and applications of AI as well as LMs for future vehicles and transportation. Based on the reviewers' feedback, as well as the evaluations of the editors, some good papers are selected for this special section. These papers, which cover broad topics, are introduced briefly as follow.

The paper "*LLM Enhanced Reconfigurable Intelligent Surface for Energy-Efficient and Reliable 6G IoV*" leverages the analytical capabilities of LLM, combined with key IoV data such as channel status, vehicle movement patterns, and quality of service requirements, to model and optimize RIS-based IoV communication systems. This work justifies the significant performance advantages of LLM-enhanced RIS, demonstrating a viable technical pathway for the development of 6G IoV.

The paper "*Large Model-assisted Federated Learning for Object Detection of Autonomous Vehicles in Edge*" proposes a novel federated learning framework that combines the strengths of both synchronous and asynchronous methods for object detection of autonomous vehicles in edge. Experimental results demonstrate this proposed method significantly enhances the model convergence and performance in object detection for autonomous vehicles.

The paper "*Optimization of Resource Allocation for V2X Security Communication based on Multi-Agent Reinforcement Learning*" proposes a resource allocation optimization framework based on the multi-agent reinforcement learning algorithm, in which a large-scale neural network model is built to train the vehicular to learn the optimal resource allocation strategy for optimal communication performance and security performance. Experimental results show that the load successful delivery rate and confidentiality performance of the vehicular communication network are effectively improved compared with some baselines while ensuring link security.

The paper "*Large Beamforming Models for Robust Wireless Transmission in Generalized Scenes of RIS-aided Intelligent IoV Network*" formulates and integrates LMs to achieve excellent hybrid beamforming within IoV-based wireless communications system which is aided by reconfigurable intelligent surfaces and base stations to provide additional channels and coverage

gains. Experimental results have consistently confirmed that this proposed LMs enabled hybrid beamforming for reconfigurable intelligent surface aided IoVs outperforms traditional schemes in the respect of system throughput and network coverage.

The paper "*PPRU: A Privacy-Preserving Reputation Updating Scheme for Cloud-Assisted Vehicular Networks*" proposes a novel privacy-preserving reputation updating scheme for cloud-assisted vehicular networks based on the elliptic curve cryptography and Paillier algorithms. Experimental results show this proposed scheme can provide strong privacy preservation, strong security, and robust reputation management with acceptable computation and communication overheads.

The paper "*Emerging Blockchain and Reputation Management in Federated Learning: Enhanced Security and Reliability for Internet of Vehicles (IoV)*" proposes a new reliable and secure federated learning for IoVs based on decentralized blockchain and reputation management. Moreover, this work uses the concept of the reputation of vehicles to select suitable participants with reliability, ultimately improving accuracy. Security analysis results, including comparisons with previous works, prove that the scheme can address security concerns.

The paper "*IoV-BERT-IDS: Hybrid Network Intrusion Detection System in IoV Using Large Language Models*" introduces a hybrid network intrusion detection system in IoVs using LLM, denoted as IoV-BERT-IDS in IoVs. Experiments demonstrate that the proposed IoV-BERT-IDS outperforms in several public datasets and shows good generalization capabilities to in-vehicle networks of different vehicles.

The paper "*CFPC: The Curbed Fake Point Collector to Pseudo-LiDAR-Based 3D Object Detection for Autonomous Vehicles*" proposes a curbed fake point collector to support 3D object detection for autonomous vehicles. The effectiveness of this proposed method is demonstrated on the public dataset. It is superior to the baseline in most situations, particularly in the categories of cars and riders.

The paper "*Large Models for Cooperative Control of Connected and Autonomous Vehicles*" proposes LMs-based approaches for cooperative control and coordination of connected and autonomous vehicle (CAV) fleets. Experimental results demonstrate enhanced traffic efficiency, safety, robustness, and scalability using LMs for cooperative control of CAV.

The paper "*Online Analytic Exemplar-Free Continual Learning with Large Models for Imbalanced Autonomous Driving Task*" proposes an analytic exemplar-free online continual learning algorithm with LMs for imbalanced autonomous driving task. Experimental results demonstrate that despite being

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an exemplar-free strategy, this proposed method outperforms various methods on the autonomous driving.

The paper “*An Efficient Division Method of Traffic Cell Based on Improved K-means Clustering Algorithm for the Location of Infrastructure in Vehicular Networks*” proposes a systematic clustering combined with genetic algorithm-K-means algorithm for traffic cell delineation method. This proposed method is validated through clustering experiments on real datasets, and its clustering advantages are demonstrated by combining clustering evaluation indicators.

The paper “*Federated Learning-based Privacy-aware Location Prediction Model for Internet of Vehicular Things*” proposes an innovative approach that leverages machine learning and edge computing techniques to predict vehicles’ subsequent locations using large-scale data, concurrently prioritizing user privacy. Furthermore, this work introduces an optimization method to monitor vehicle speed and outlines a comprehensive attack model, along with effective protection measures.

The paper “*Federated Reinforcement Learning-Empowered Task Offloading for Large Models in Vehicular Edge Computing*” proposes a federated learning framework to judiciously determine the proportion of resource allocation of auxiliary vehicles through the implementation of inventive mechanisms. Experimental results validate this proposed method not only leads to the improvement of key system performance indicators, but also ensures the comprehensive exploitation of the computing resources of mobile vehicles.

The paper “*Integration of LLM and Human-AI Coordination for Power Dispatching with Connected Electric Vehicles under SAGVNs*” proposes a framework LLM-D3PG, LLM-guided dual Deep Deterministic Policy Gradient, which empowers electric vehicle drivers to articulate their expectations for AI-generated dispatching strategies using natural language instructions. Experimental results demonstrate the effectiveness of the proposed framework and show outperformance in comparison to some existing methods for power dispatching tasks with the random connection of electric vehicles in space-air-ground integrated vehicular networks.

The paper “*A Novel Design and Comparison for Connected Autonomous Vehicle Enabled Signal-Free Intersection Control Scheme*” models the operations of three unconventional autonomous intersections and compare their operational performance with one conventional autonomous intersection. Results show that unconventional vehicle crossing paths at intersections are likely to reduce vehicle delay and fuel consumption compared with conventional intersections in most scenarios.

The paper “*Superpixel Prior Cluster-level Contrastive Clustering Network for Large-scale Urban Hyperspectral Images and Vehicle Detection*” proposes a superpixel prior cluster-level contrastive clustering network for large-scale hyperspectral image clustering tasks. Experimental results indicate this proposed method performs well in both hyperspectral image clustering and vehicle recognition tasks.

The paper “*AACS: A Secure Attribute Based Data Access Control for Cloud-aided Internet of Vehicles Using SGX*” develops a pure cryptographic construction to enforce attribute-based access control model in untrusted cloud for IoVs based on ciphertext policy-attribute based encryption and explore three limitations. Experimental results show this method can effectively

lower-bound the costs that would be incurred when using more advanced and more expensive predicate encryption schemes.

The paper “*Block-FeDL: Electric Vehicle Charging Load Forecasting using Federated Learning and Blockchain*” proposes Block-FeDL, a blockchain-based federated learning approach for electric vehicle charging load forecasting considering the private and sensitive charging information of each user. Experimental results show that the Block-FeDL outperforms the second-best baseline method obviously in terms of mean square error, mean absolute error, and root mean square error for forecasting the electric vehicle charging load.

The paper “*An Efficient 6G Federated Learning-enabled Energy-Efficient Scheme for UAV Deployment*” proposes a multi-UAV energy efficient coverage deployment algorithm based on a spatial adaptive play. Experimental results show this proposed method outperforms all the state-of-the-art algorithms by achieving an improved data rate and balanced SINR value.

The paper “*Intelligent Prediction of Urban Road Network Carrying Capacity and Traffic Flow Based on Deep Learning*” proposes a spatiotemporal dynamic deep network considering the spatial characteristics of urban road networks and the temporal information of traffic speeds. Experimental results have demonstrated the effectiveness and superiority of this proposed model.

The paper “*Learning-based Cooperative Cache Strategy Integrating Content Recommendation in IoV*” proposes a clustering caching model by considering constraints for tolerable delay of tasks and the residence time of vehicles. Experimental results demonstrate this proposed method has significant advantages in saving task delay and system energy consumption compared to other solutions.

The paper “*PR-OppCL: Privacy-Preserving Reputation-Based Opportunistic Federated Learning in Intelligent Transportation System*” proposes an innovative framework for privacy-preserving reputation-based opportunistic federated learning to address privacy concerns in intelligent transportation systems. Finally, experimental results demonstrate a significant reduction in training rounds and total time, outperforming existing methods.

The paper “*Electromagnetic Exposure Safety Assessment of Medical Implants in the Human Body for Electric Vehicle Wireless Power Transfer Considering Uncertainty*” analyzes the safety issues of human electromagnetic exposure containing implants in the electric vehicle wireless power transfer. Experimental results show this scheme can obtain accurate results with fewer training sets and have higher computational efficiency.

The paper “*Reliable Computation Offloading of DAG Applications in Internet of Vehicles Based on Deep Reinforcement Learning*” proposes a partial offloading algorithm for DAG application via deep reinforcement learning in IoVs. Experimental results show this proposed method achieves better comprehensive utility than the baseline algorithms.

The paper “*Transportation Route Optimization of Municipal Solid Waste based on Improved Ant Colony Algorithm in Internet of Vehicles*” proposes an enhanced ant colony optimization algorithm for transportation route optimization of municipal solid waste in IoVs. Experimental findings reveal that, compared with both separate and unified schemes, this proposed model yields reductions in collection and transportation costs.

The paper “*Location Optimization Enhancement of Wireless Sensor Networks in Intelligent Transportation System Based on Reverse Elite Large Model*” proposes a multi-objective whale optimization algorithm based on reverse elite large model for location optimization in intelligent transportation systems. Experimental results show this proposed method can provide higher-quality non-dominated solutions, select suitable node locations, and optimize their performance metrics.

The paper “*Digital Twin-Aided Vehicular Edge Network: A Large-Scale Model Optimization by Quantum-DRL*” introduces a comprehensive model that integrates quantum computing with a deep reinforcement learning model for a digital twin-aided vehicular edge network. Experimental results demonstrate notable improvements in task offloading efficiency, highlighting the model’s efficacy over conventional deep reinforcement learning models.

The paper “*Multimodal 3D Object Detection Based on Sparse Interaction in Internet of Vehicles*” proposes the multimodal grid encoder with positional information, through positional information implicitly guiding and the transformer-based attention mechanism for grid-level features, improves the perception of detection scene context spatial information and enhances the robustness of the algorithm. Experimental results demonstrate this proposed algorithm has superior 3D object detection performance in Internet of Vehicles (IoVs).

The paper “*Attribute-Consistency Reversible Pedestrian De-Identification in Intelligent Transportation*” proposes an Attribute-Consistency Reversible Pedestrian De-Identification (ACRPD) framework for intelligent transportation and pedestrian navigation scenarios. Experimental results conducted on two datasets demonstrate the effectiveness of the proposed ACRPD method.

The paper “*Intelligent Transportation Vehicle Road Collaboration and Task Scheduling Based on Deep Learning in Augmented Internet of Things*” proposes an intelligent transportation vehicle road collaboration and task scheduling method based on deep learning in Internet of Things. Experimental results show this proposed method can effectively reduce the overall cost of edge cloud computing, and improve the safety and efficiency of intelligent driving.

The paper “*A New Edge Computing Method Based on Deep Learning in Intelligent Transportation System*”, combines deep learning with edge computing to take traffic flow detection in intelligent transportation system as the research base point. Finally, the effectiveness and superiority of this proposed method are verified by comparison experiments with various prediction models.

The paper “*Adaptive Spatio-Temporal Relation based Transformer for Traffic Flow Prediction*” introduces the ASTRformer, an innovative approach to traffic flow prediction that emphasizes the fusion of spatial and temporal information in historical data through an adaptive spatiotemporal relation learning mechanism. Experimental results on six datasets demonstrate this proposed method effectively captures spatiotemporal dependencies, achieving state-of-the-art performance across various prediction metrics.

The paper “*Multiple Models-Based Fault Tolerant Control of Levitation Module of Maglev Vehicles against Partial Actuator Failures*” proposes a fault-tolerant control strategy to solve the problem of partial actuator failures in the levitation module control system of maglev vehicles based on multi-models. The effectiveness of this proposed method with actuator fault compensation technology for multiple-point levitation modules of maglev vehicles is verified.

The paper “*DMDAT: Diffusion Model-based Data Augmentation Technique for Vision-based Accident Detection in Vehicular Networks*” presents a diffusion model-based data augmentation for enhanced vehicular network applications. This work uses a vision-based accident detection dataset and a MobileNet-based deep learning model to understand the efficacy of the proposed data augmentation technique. Experimental results indicate this proposed data augmentation technique improves the performance notably.

The paper “*Intelligent Optimization Algorithm for Green IoV Networks based on SSA*” states power consumption optimization plays an important role in green IoVs. In this paper, the author investigates the intelligent power allocation optimization of green cooperative IoVs based on SSA. Experimental results show this proposed method can achieve a shorter running time and a good outage probability performance, thereby enhancing the system’s efficiency.

All papers tackle different but extremely relevant domain vectors of AI and LMs for future vehicles and transportation. Finally, we would like to extend our gratitude to all the authors who submitted their research to this special section. We also acknowledge the invaluable contributions of numerous field experts who participated in the review process and offered insightful suggestions to enhance the content and presentation. Special thanks go to Professor Abbas Jamalipour, the Editor-in-Chief, and the publishing team for their support and valuable feedback during the final stages of completing this special section.

SHAHID MUMTAZ, *Guest Editor*
Nottingham Trent University
NG11 8NS Nottingham, U.K.
dr.shahid.mumtaz@ieee.org

BO AI, *Guest Editor*
Beijing Jiaotong University
Beijing 100044, China
boai@bjtu.edu.cn

IKRAM ASHRAF, *Guest Editor*
Nokia
90014 Oulu, Finland
ikram.ashraf@nokia-bell-labs.com

ANWER AL-DULAIMI, *Guest Editor*
EXFO Inc.
Québec City, G1M 2K2 Canada
anwer.al-dulaimi@exfo.com